

Lab 23 - Expected verification output

Run from the lab directory:

```
bash verification/checks.sh
```

Honesty note

checks.sh is read-only and queries your live cluster. It auto-detects whether you installed metrics-server (Part 5 Path A) or skipped it (Path B).

Path B is a valid completion of this lab. Its metrics checks report SKIP, not FAIL, and the script still exits 0. A stock kind cluster has no metrics-server, and pretending otherwise would be the exact dishonesty this lab teaches you to avoid.

Path B - metrics-server NOT installed (default on stock kind)

```
== Namespace isolation
PASS namespace kcna-lab23 exists
PASS Pod Security Admission enforce=baseline

== Part 1 - access-log corpus loaded from data/
PASS configmap/depot-access-log exists
PASS corpus loaded (30 lines)
PASS corpus contains the ERROR/upstream_timeout records used in Part 4

== Part 2 - event-producing workloads
PASS deployment/depot-portal exists
PASS deployment/gate-scanner exists
PASS pod/depot-gate-agent exists (the FailedMount event source)
PASS depot-portal Pod has 2 containers (portal + audit-sidecar)

== Part 3 - Events and field selectors
PASS compound field selector returns 2 Warning event(s) about Pods
PASS reason=FailedMount selector returns the depot-gate-agent event
PASS non-indexed field selector is rejected server-side (expected)

== Part 4 - logs
PASS found depot-portal pod depot-portal-58d6c47f9b-rz8nq
PASS container 'portal' is emitting the data/ corpus to stdout
PASS container 'audit-sidecar' is shipping its file to stdout
PASS gate-scanner has restarted 4 time(s) - --previous is meaningful

== Part 5 - metrics pipeline (Path A or Path B)
SKIP metrics.k8s.io not installed - Path B. This is a VALID completion.
SKIP kubectl top checks not applicable on Path B
PASS kubectl top fails with the documented Path B error (expected on stock kind)

== Worksheet
PASS worksheet present at data/event-triage-worksheet.csv
PASS worksheet header intact

== Blast-radius guard
PASS no lab-23 workloads/config in kube-system
SKIP no kube-system RoleBinding (Path B, or already cleaned up)

SUMMARY: 18 passed, 0 failed, 4 skipped
```

Exit code 0.

Path A - metrics-server installed

The Part 5 and blast-radius blocks change to:

```
== Part 5 - metrics pipeline (Path A or Path B)
PASS APIService v1beta1.metrics.k8s.io exists (Path A)
PASS aggregation layer reports Available=True
PASS --kubelet-insecure-tls present (required on kind; lab clusters only)
PASS metrics-server image is pinned (registry.k8s.io/metrics-server/metrics-server:v0.7.2)
PASS kubectl top pods returns data

== Blast-radius guard
PASS no lab-23 workloads/config in kube-system
PASS the one documented kube-system RoleBinding is present (Path A)

SUMMARY: 24 passed, 0 failed, 0 skipped
```

Exit code 0.

Note that the script asserts kubectl top pods returns data - it does not

assert any particular millicore or byte value, because no such value can be

predicted. See Step 5.6 in README.md for the one quantitative claim that

is defensible: the indexer container is capped by a 100m CFS quota and

must never be reported above it.

Common partial runs

Run too early, before events have accumulated

```
== Part 3 - Events and field selectors
FAIL no Warning events yet - wait ~60s after Step 2.1, then re-run
```

Exit code 1. Not a defect - BackOff needs a restart cycle and FailedMount is re-raised on a timer. Wait 60 seconds and re-run.

Path A, run inside the first scrape window

```
FAIL kubectl top pods still failing - allow ~60s for the first scrape window
```

metrics-server needs at least one --metric-resolution=15s cycle after becoming Ready. Wait and re-run. If it never clears, check kubectl logs -n kcna-lab23 deploy/metrics-server against the Troubleshooting table in README.md.

Path A, RoleBinding not applied

```
FAIL APIService Available='False' - see the Troubleshooting table
SKIP no kube-system RoleBinding (Path B, or already cleaned up)
```

metrics-server cannot read the extension-apiserver-authentication ConfigMap. Either apply manifests/91-metrics-server-kube-system-rolebinding.yaml or switch to Path B.

Manual observations the script cannot make for you

#	Observation	Where
1	Unsorted kubectl get events output is not chronological	Step 3.1 vs 3.2
2	An event showing (x2 over 3m1s) - aggregation, not enumeration	Step 3.3
3	count, firstTimestamp and lastTimestamp on the raw Event object	Step 3.8
4	--event-ttl is not set by kubeadm, so the 1 hour default applies	Step 3.9
5	--previous showing the FATAL line that the current container has not reached	Step 4.5
6	The audit file existing inside the container and reaching kubectl logs only because a tail -f re-emits it	Step 4.7
7	kubectl logs returning NotFound immediately after the Pod is deleted	Step 4.9
8	The two *different* metrics failure strings: Metrics API not available vs ServiceUnavailable	Steps 5.1 and 6.2